

Performance Statistics

Rebuilding Time

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When conducting Management Strategy Evaluation objective need to be agreed and then quantified in the form of statistics known as **Performance Statistics**.

Ensuring sustainability requires preventing the stock from becoming overfished, so there is a low probability of compromising productivity. Therefore, many fishery management bodies define a biomass limit reference (B_{lim}) above which the stock should be maintained with high probability. Biomass limits should be based exclusively on the biology of the stock and its resilience to fishing pressure. They should not consider economic factors, as the objective is to reduce risks from a biological perspective, and must integrate dynamic processes such as growth, fecundity, recruitment, mortality and connectivity into indices for exploitation level and spawning reproductive potential.

Some remaining issues relate to how to model management objectives. For example, B_{lim} is used as a limit corresponding to a biomass at which recruitment is impaired. This is commonly estimated from the stock recruitment relationship, e.g. by the break point of a segmented regression. The intention is to prevent recruitment overfishing, the undesirable state in which adults of a species are so overfished that they cannot reproduce fast enough to replenish the stock. This is problematic since there may be no evidence of a stock-recruitment relationship if a stock has not been overfished, recruitment may be influenced by environmental drivers independently of SSB, the occurrence of maternal effects may mean that total biomass is not a good measure of spawning reproductive potential (Kell et al, 2016). The reduction of the effect of density dependence at low stock size (compensation) means that as biomass declines recruitment per capita recruitment actually increases. While in the case of depensation recruitment is reduced at lower biomasses. Therefore, recruitment declines proportionally less than SSB. However, productivity, is not only determined by recruitment but also by natural mortality, growth and fecundity-at-age, which may also be impacted by density dependent processes (Mangel2013?).

A way of doing this consistently across stocks and model assumptions is to calculate the rebuilding relative to generation time for a depleted stock to a target level, e.g. the biomass that will support the maximum sustainable yield (B_{MSY}). For example as required for depleted fisheries in the United States. This is also consistent with the IUCN Red List of Threatened Species (Hilton-Taylor and Brackett 2000), which uses generation time as a measure of population decline, as it allows the same criteria to be used for species with extremely different life spans (Mace et al. 2008).

Therefore, as a performance statistic for safety we propose the biomass from which a stock can recover to B_{MSY} within a given time scale and level of fishing mortality ($B_{rebuild}$). The details of which determine the level of risk and so are a management decision. For example, if the management objective is that a depleted stock should recover within a generation time with zero fishing, the corresponding depletion level provides the limit reference point.

In an MSE the 5% level could be used for evaluation, i.e. a strategy is only successful if it keeps a stock above $B_{rebuild}$ with a 95% probability.

Example using ‘FLBRP’

For a stock the ‘FLBRP’ object can be derived, this holds the expected values for the mortality, selectivity, mass, maturity-at-age, and the stock recruitment relationship, and can be used to calculate the expected, i.e. equilibrium values of biomass for any given level of fishing mortality.

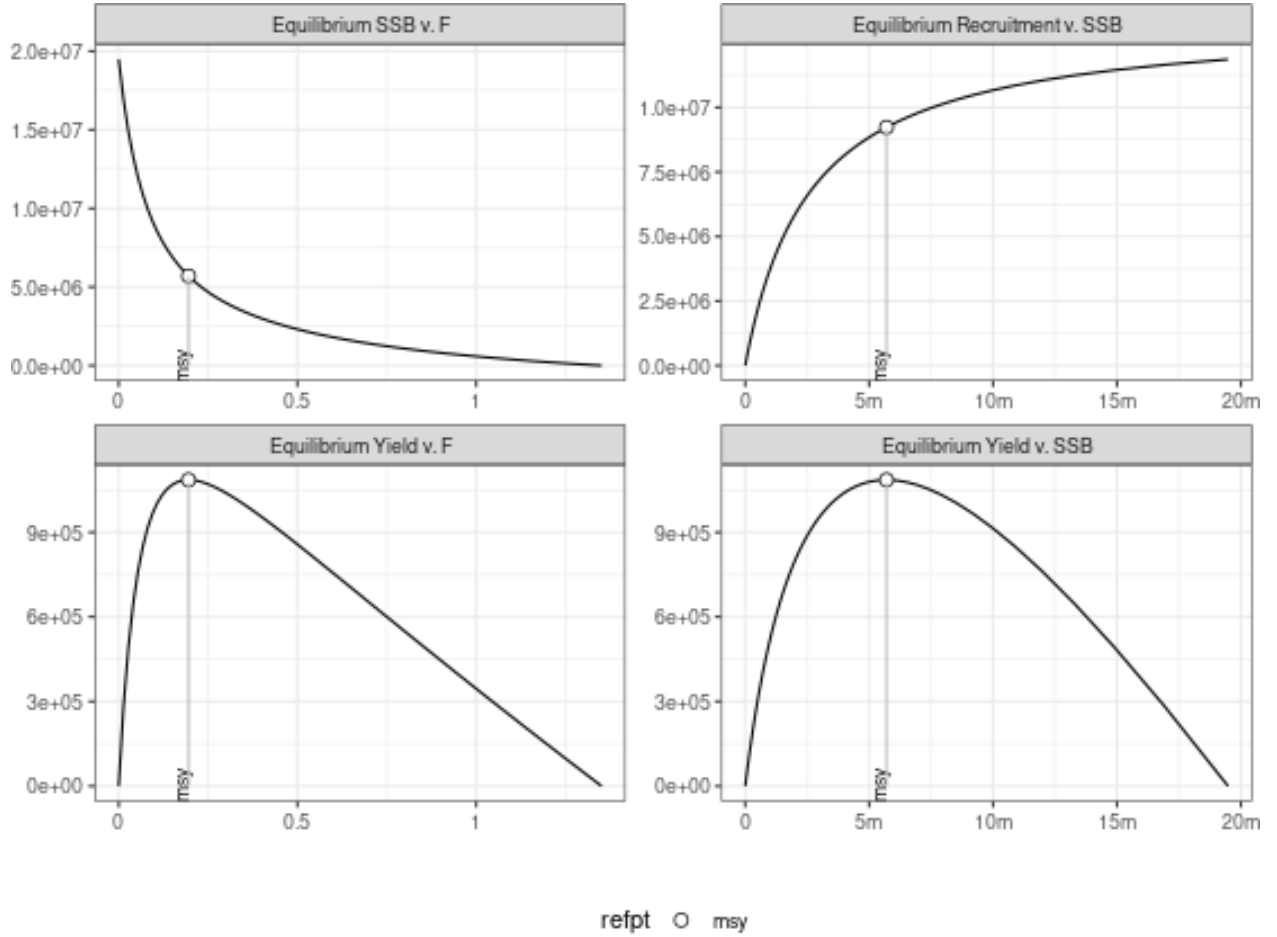


Figure 1 Equilibrium curves showing estimates of SSB, recruitment yield and F.

The ‘FLBRP’ object is then used to simulate ‘FLStock’ objects that can be projected through time for a given level of fishing mortality. In **Figure 2** the stock is initially at different levels of depletion and then projected for $F = 0$. The stock then recovers, the dashed horizontal line is SSB/B_{MSY} , and the green vertical line is half a generation time, and the red vertical line one generation time. For a desired time for recovery the level that the stock should not fall below can be read from the curves, and is summarised in **Figure 3**.

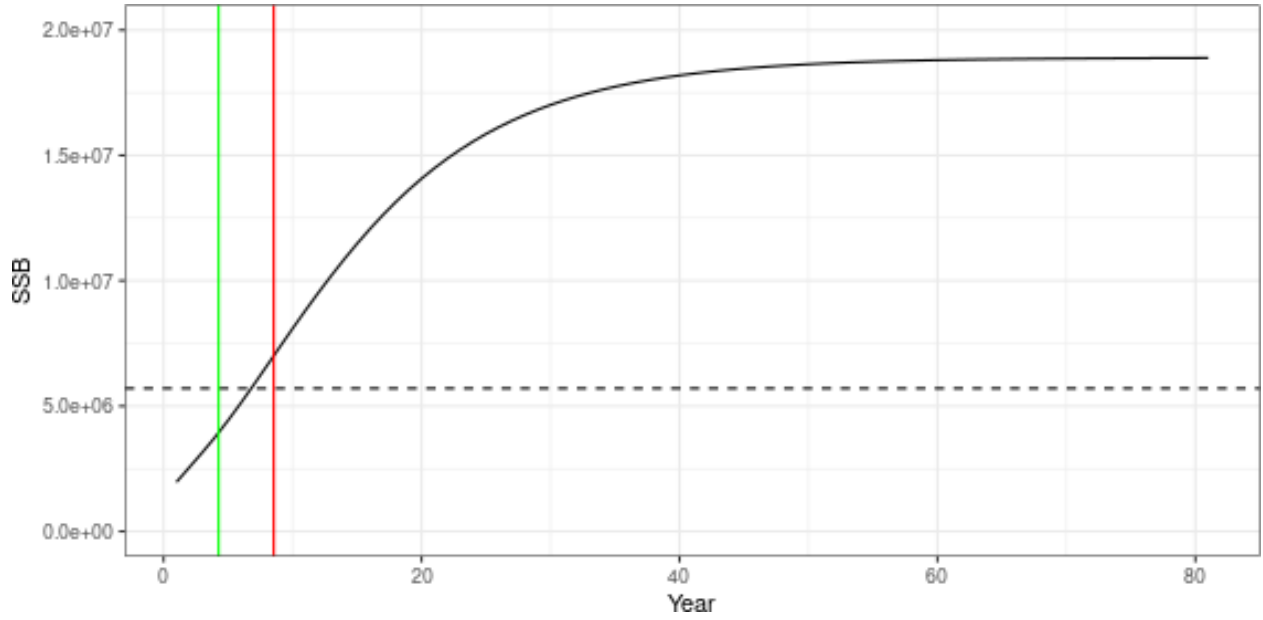


Figure 2 Recovery from depletion, vertical lines are half (green) and one (red) generation time

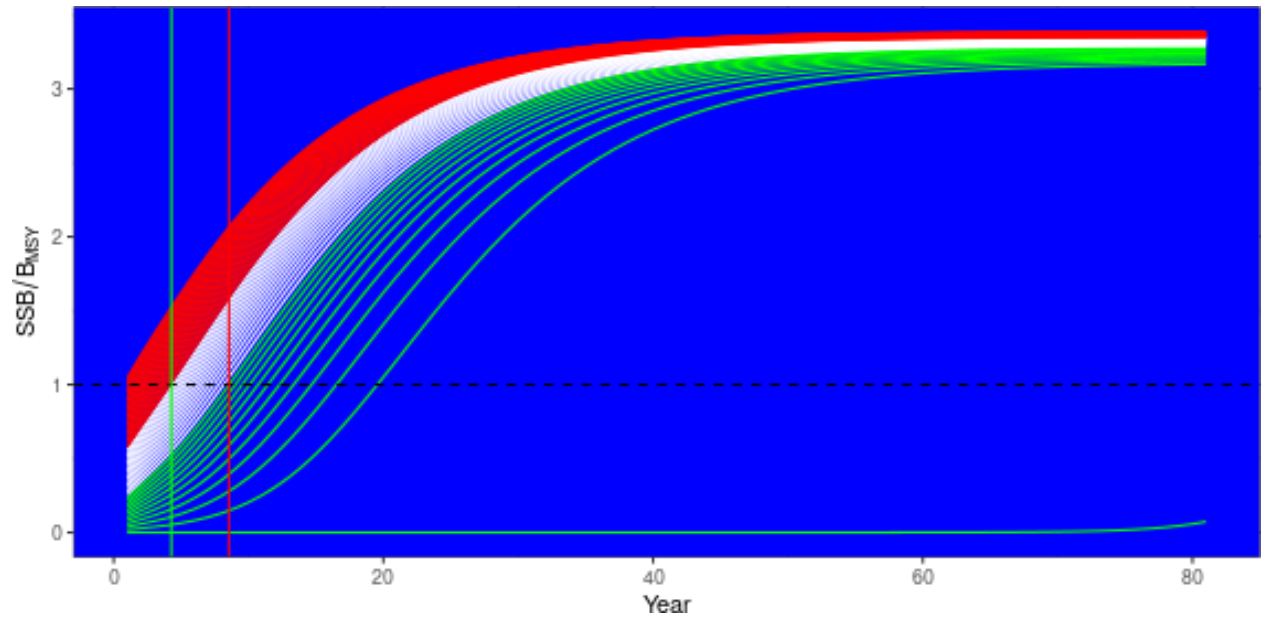


Figure 3 Recovery from different depletion levels, red horizontal line is B_{MSY} , vertical lines are half (green) and one (red) generation time

If the management objective is in the case of stock depletion is to close the fishery to recover a stock in a generation time then the biomass limit that should be avoided with high probability is 25% of B_{MSY} . If fishing is still to be allowed under recovery then the biomass limit will have to be higher.

As a performance statistic you could then say there must be a 95% prob that you can rebuild within a given fraction of a generation time with $F=0$.

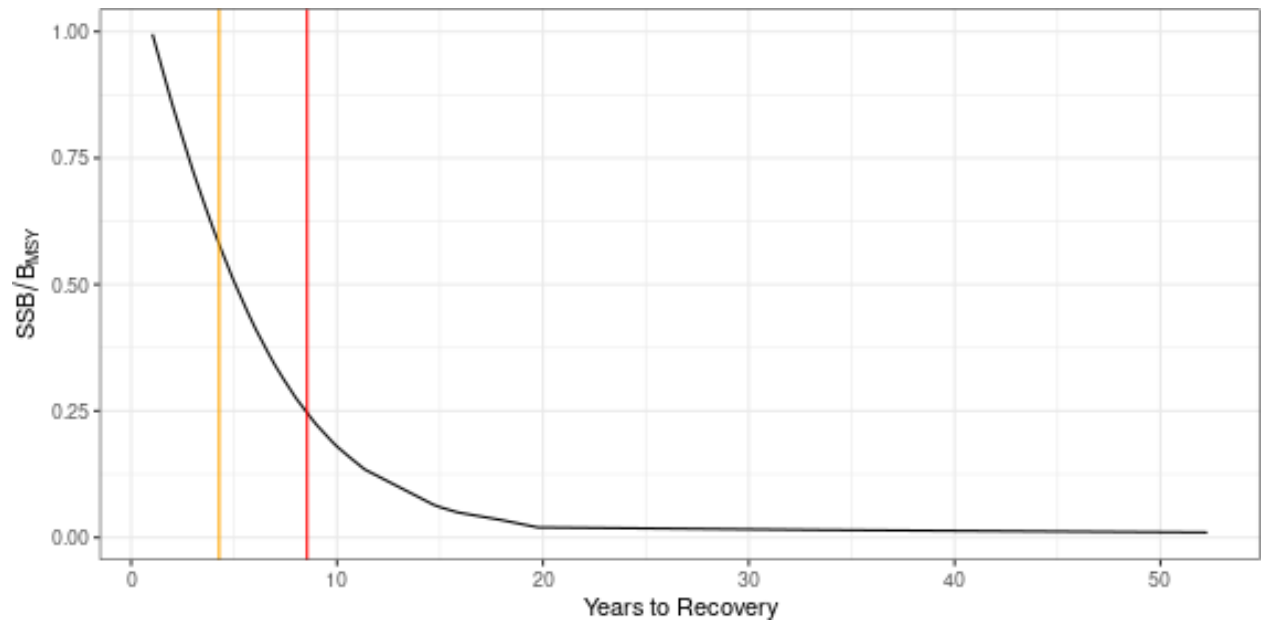
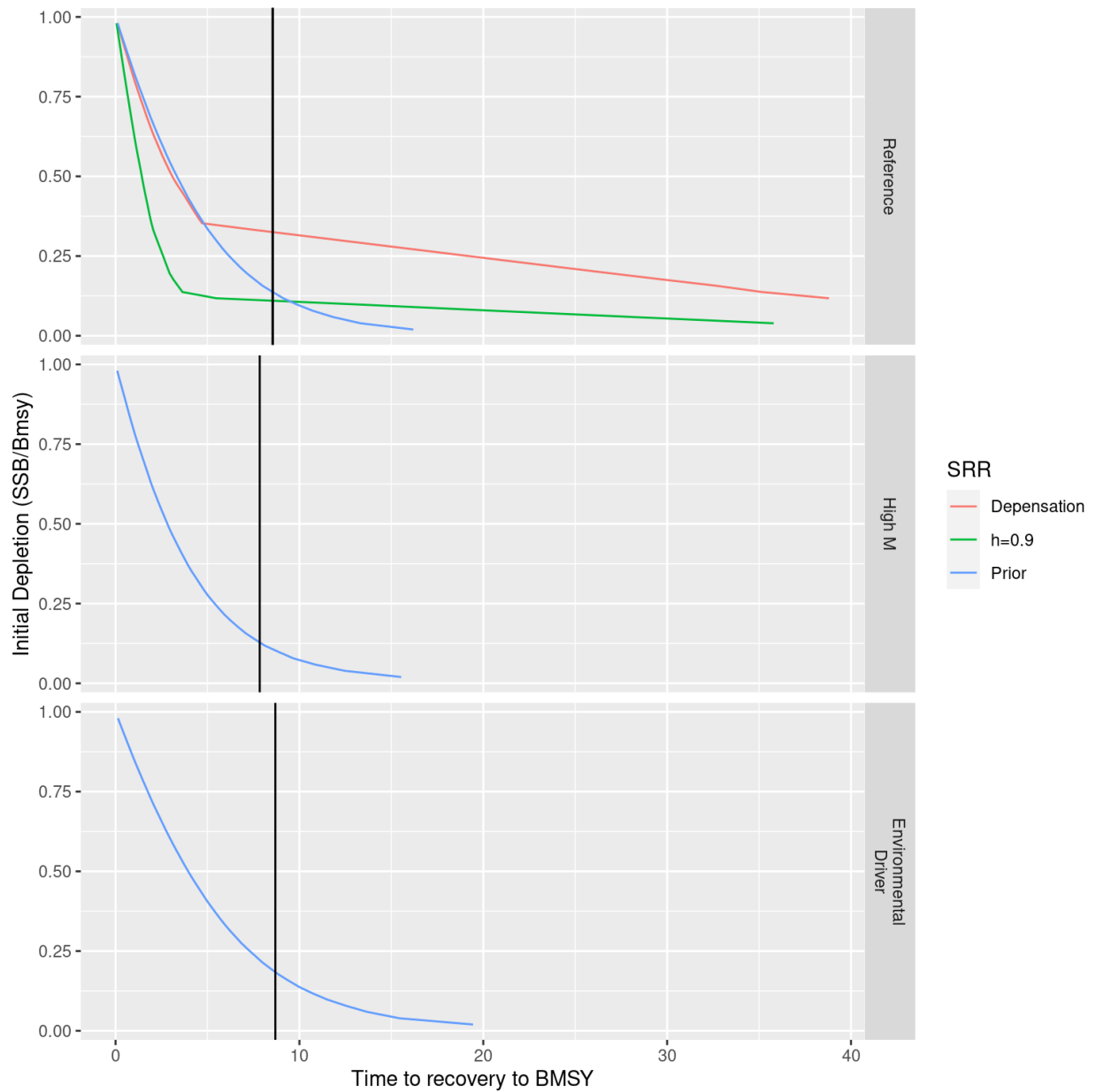


Figure 4 Time to recover from different depletion levels, vertical lines are half (white) and one (red) generation time

Impact of M and the SRR



0:	19.965325:	10.1229	-0.916291	1.91288
1:	19.795713:	10.1627	-0.888221	1.91357
2:	19.768537:	10.1484	-0.872385	1.91251
3:	19.763467:	10.1551	-0.866864	1.91226
4:	19.762547:	10.1525	-0.863914	1.91168
5:	19.762314:	10.1544	-0.862401	1.90855
6:	19.762268:	10.1535	-0.862582	1.90468
7:	19.762238:	10.1533	-0.862163	1.90657
8:	19.762229:	10.1536	-0.862038	1.90713
9:	19.762228:	10.1536	-0.862076	1.90702
10:	19.762228:	10.1536	-0.862075	1.90702
0:	90.016145:	8.77345	-0.916291	1.05279

1:	43.892164:	9.28078	-0.0695157	1.21276
2:	43.836695:	10.1209	-0.609402	1.26451
3:	38.774357:	9.98359	-0.270550	0.923462
4:	37.683715:	10.0659	-0.444244	0.840813
5:	37.617093:	10.0893	-0.456290	0.891391
6:	37.608816:	10.1425	-0.454067	0.870829
7:	37.607875:	10.1336	-0.451616	0.868423
8:	37.607860:	10.1338	-0.452187	0.868684
9:	37.607860:	10.1337	-0.452150	0.868735
10:	37.607860:	10.1337	-0.452152	0.868730
0:	98.134697:	17.0444	-0.916291	-0.105123
1:	66.204427:	17.6782	-0.215462	0.222216
2:	57.386679:	17.6611	-0.438680	0.222004
3:	54.373848:	17.5955	-0.880259	0.256308
4:	52.182700:	17.4034	-0.731071	0.632218
5:	51.700487:	17.4600	-0.750783	0.597805
6:	51.419157:	17.4745	-0.755889	0.530429
7:	51.327330:	17.5123	-0.737745	0.475505
8:	51.309615:	17.5019	-0.747032	0.467676
9:	51.308954:	17.5046	-0.745703	0.465233
10:	51.308767:	17.5057	-0.747009	0.461721
11:	51.308646:	17.5052	-0.745795	0.461631
12:	51.308645:	17.5052	-0.745841	0.461592
13:	51.308645:	17.5052	-0.745839	0.461601
14:	51.308645:	17.5052	-0.745840	0.461602
0:	10.662667:	15.2814	-0.916291	0.215888
1:	1.8827555:	15.5267	-1.04446	0.263769
2:	-4.1584956:	15.5871	-1.31877	0.267349
3:	-5.7707734:	15.5216	-1.51201	0.222162
4:	-6.5514620:	15.6478	-1.53372	0.0570197
5:	-6.6401656:	15.6274	-1.55941	0.0850229
6:	-6.6502214:	15.6305	-1.57499	0.0891227
7:	-6.6518858:	15.6260	-1.57609	0.0862232
8:	-6.6531418:	15.6255	-1.57522	0.0915800
9:	-6.6532200:	15.6231	-1.57467	0.0964097
10:	-6.6533343:	15.6237	-1.57727	0.0978175
11:	-6.6534284:	15.6238	-1.57619	0.0964717
12:	-6.6534284:	15.6238	-1.57620	0.0965118
13:	-6.6534284:	15.6238	-1.57620	0.0965103
0:	51.944736:	11.4870	-0.916291	2.14391
1:	50.127249:	11.6258	-0.652428	2.14344
2:	48.885818:	11.5454	-0.682422	2.13793
3:	48.877253:	11.5051	-0.758061	2.13132
4:	48.730090:	11.5422	-0.736375	2.13089
5:	48.714232:	11.5275	-0.730490	2.12825
6:	48.708318:	11.5342	-0.726225	2.11430
7:	48.706688:	11.5329	-0.731964	2.09940
8:	48.706101:	11.5337	-0.728306	2.09785
9:	48.706056:	11.5334	-0.728784	2.09785
10:	48.706054:	11.5336	-0.729328	2.09789
11:	48.706049:	11.5334	-0.729098	2.09831
12:	48.706049:	11.5335	-0.729057	2.09788
13:	48.706049:	11.5335	-0.729083	2.09810
14:	48.706049:	11.5335	-0.729087	2.09810

15:	48.706049:	11.5335	-0.729088	2.09808
16:	48.706049:	11.5335	-0.729087	2.09807
0:	70.559422:	16.4650	-0.916291	0.901720
1:	50.778048:	16.6680	0.0605462	0.834096
2:	47.838766:	16.8101	-0.143124	0.656977
3:	45.887963:	16.6411	-0.396811	0.645757
4:	45.688468:	16.5973	-0.305013	0.631216
5:	45.653110:	16.5938	-0.337431	0.618891
6:	45.652448:	16.6039	-0.333284	0.622062
7:	45.651859:	16.5999	-0.336147	0.632283
8:	45.651660:	16.5978	-0.334537	0.628430
9:	45.651648:	16.5991	-0.335967	0.625163
10:	45.651619:	16.5995	-0.334981	0.627567
11:	45.651610:	16.5989	-0.335140	0.627416
12:	45.651609:	16.5990	-0.335183	0.627222
13:	45.651609:	16.5990	-0.335173	0.627256
14:	45.651609:	16.5990	-0.335173	0.627254
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1:	43.499074:	15.0945	-0.706194	1.28918
2:	41.716672:	14.9820	-0.716358	1.26789
3:	41.568918:	14.9501	-0.746679	1.25951
4:	41.559130:	14.9588	-0.752947	1.25724
5:	41.556354:	14.9545	-0.756724	1.24793
6:	41.551653:	14.9616	-0.754372	1.23998
7:	41.549079:	14.9587	-0.754459	1.22944
8:	41.546953:	14.9634	-0.754107	1.21955
9:	41.546443:	14.9639	-0.756277	1.20883
10:	41.546367:	14.9642	-0.755346	1.20899
11:	41.546356:	14.9640	-0.755282	1.20999
12:	41.546355:	14.9639	-0.755179	1.21025
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15:	41.546354:	14.9639	-0.755222	1.21034
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1:	90.874021:	11.8377	0.0103326	0.334835
2:	83.915219:	12.4485	-0.254936	1.08088
3:	82.438031:	12.5912	-0.0821163	0.683385
4:	82.122344:	12.5948	-0.155777	0.720669
5:	82.100766:	12.5345	-0.155579	0.765341
6:	82.098312:	12.5181	-0.160448	0.767942
7:	82.098247:	12.5178	-0.159771	0.766545
8:	82.098246:	12.5178	-0.159745	0.766449
9:	82.098246:	12.5178	-0.159745	0.766450
0:	78.461633:	16.5552	-0.916291	-0.261546
1:	45.668462:	16.9460	-0.0101555	-0.0998106
2:	42.434501:	17.2165	-0.251476	-0.0316525
3:	42.044121:	17.1927	-0.373741	-0.0920500
4:	41.968374:	17.2318	-0.315044	-0.211159
5:	41.942150:	17.2671	-0.342571	-0.201795
6:	41.937215:	17.2478	-0.338028	-0.177828
7:	41.936904:	17.2294	-0.338237	-0.178239
8:	41.936377:	17.2390	-0.336990	-0.180031
9:	41.936372:	17.2386	-0.337363	-0.179497
10:	41.936372:	17.2386	-0.337295	-0.179580

11: 41.936372: 17.2386 -0.337295 -0.179580

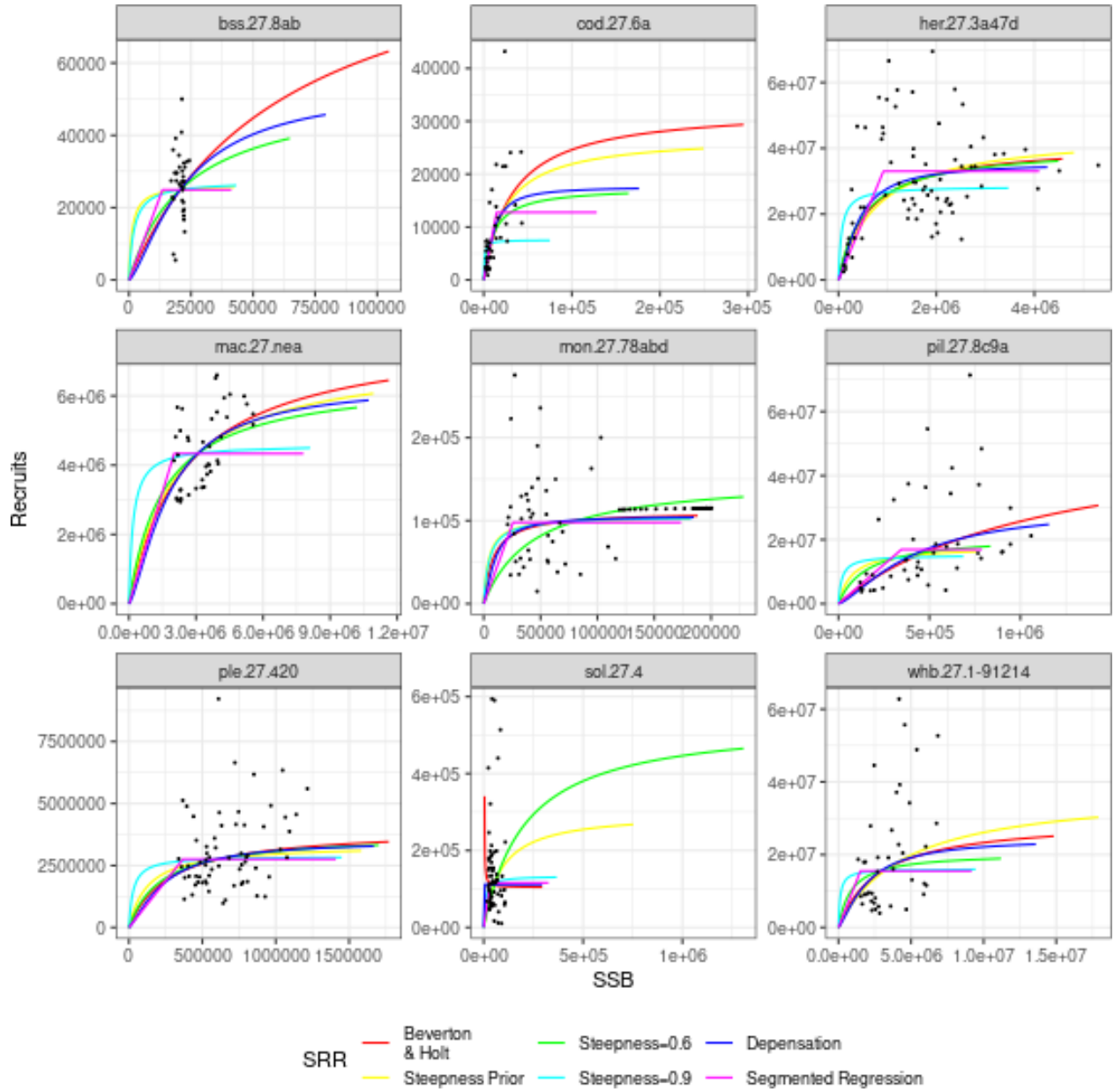


Figure 5

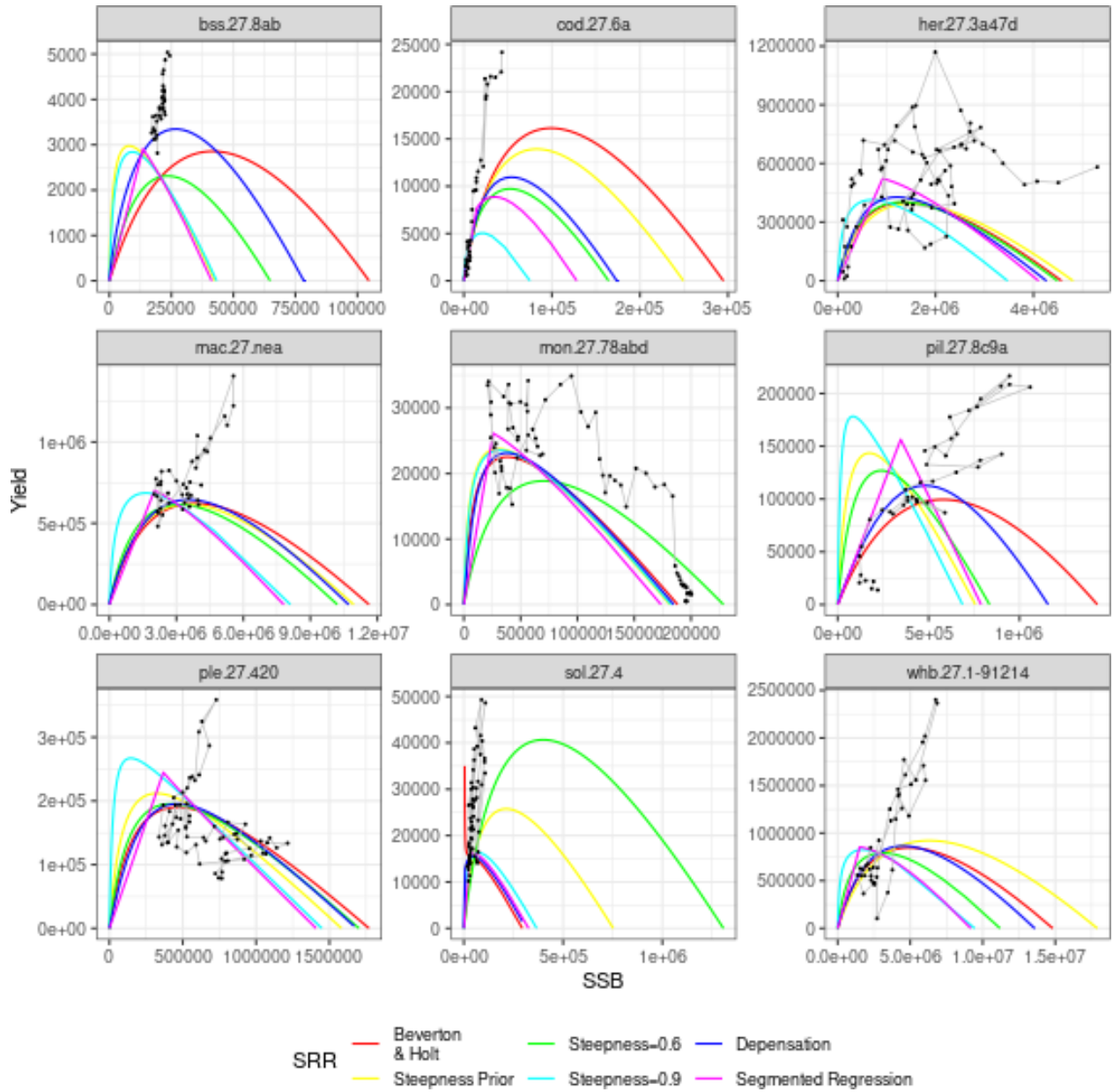


Figure 6

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Error in h(simpleError(msg, call)) :
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Error in h(simpleError(msg, call)) :
  error in evaluating the argument 'object' in selecting a method for function 'ssb': error in evaluating
Error in h(simpleError(msg, call)) :
  error in evaluating the argument 'object' in selecting a method for function 'ssb': error in evaluating
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  error in evaluating the argument 'x' in selecting a method for function 'as.data.frame': missing value

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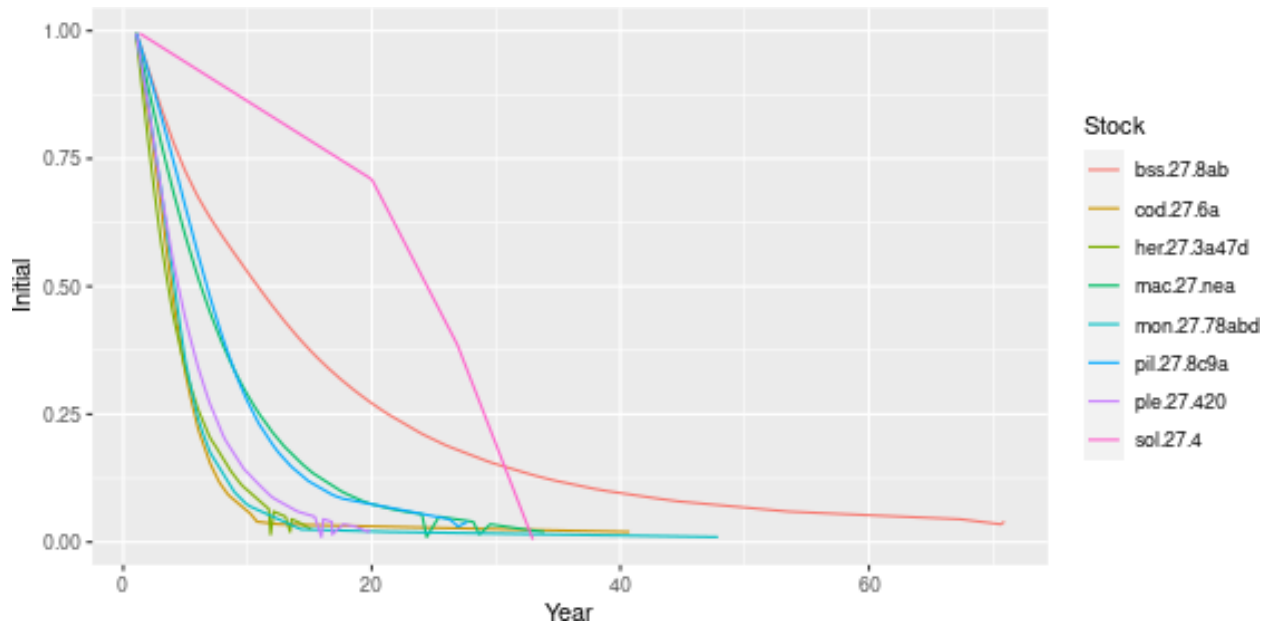


Table 1: Table 1: Ratio between ICES and scenario B_{MSY}

.id	Beverton & Holt	Steepness Prior	Steepness=0.6	Steepness=0.9	Depensation
bss.27.8ab	0.455	2.320	0.811	2.068	0.688
cod.27.6a	0.381	0.440	NA	1.565	0.696
her.27.3a47d	1.173	1.063	1.192	2.252	1.318
mac.27.nea	0.692	0.751	0.854	NA	0.726
mon.27.78abd	0.987	1.225	0.532	1.155	1.035
pil.27.8c9a	1.172	3.949	2.838	8.067	1.413
ple.27.420	1.155	1.610	1.261	3.570	1.125
sol.27.4	NA	0.341	0.180	0.956	1.526
whb.27.1-91214	0.682	0.544	1.125	2.096	NA

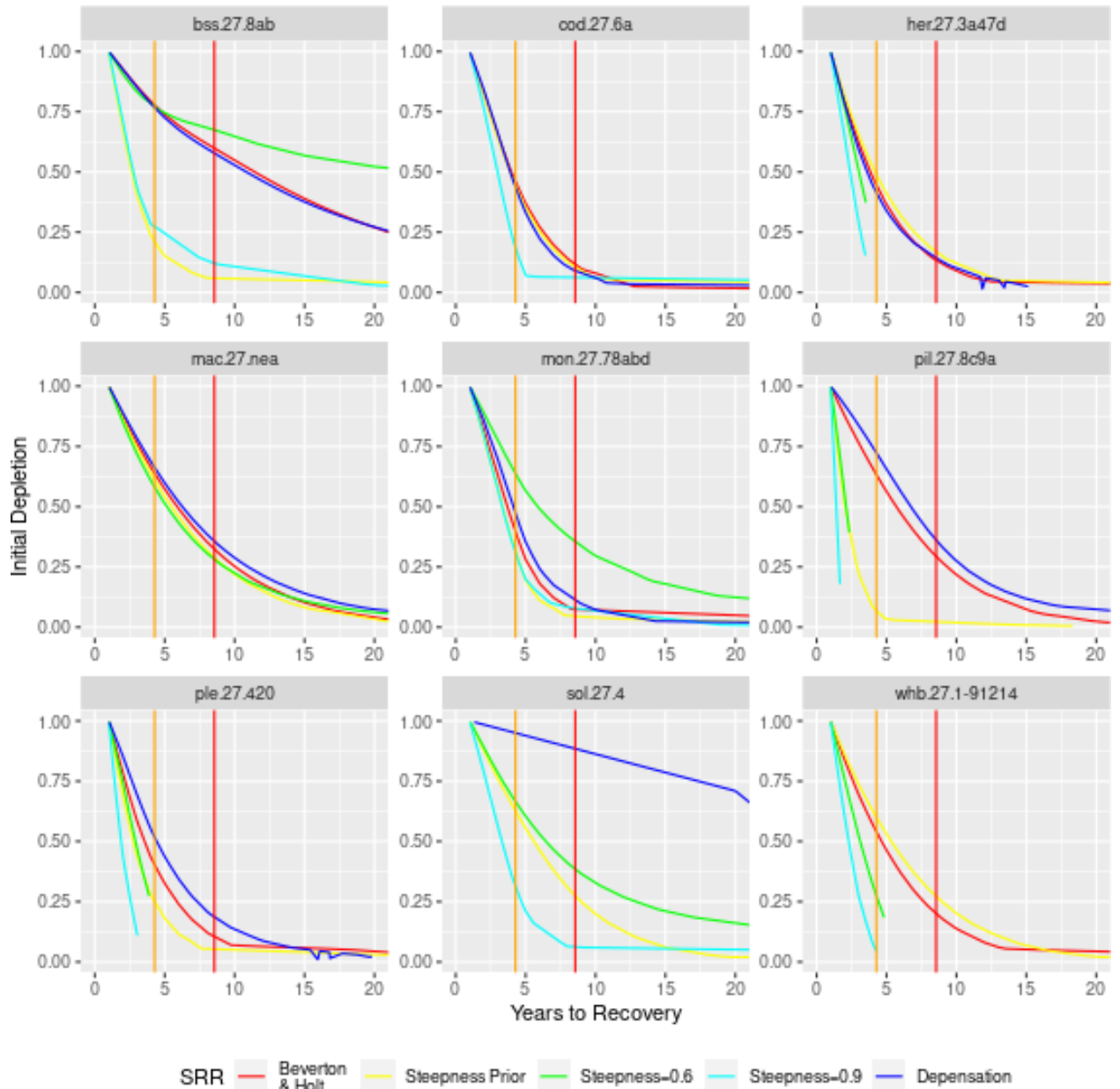


Table 2: Table 2: Rebuilding Time

.id	Beverton & Holt	Steepness Prior	Steepness=0.6	Steepness=0.9	Depensation
bss.27.8ab	19.261	-0.365	22.261	-0.013	12.976
cod.27.6a	7.881	7.181	NA	2.606	5.560
her.27.3a47d	2.868	3.515	2.435	0.340	2.308
mac.27.nea	5.609	4.735	3.662	NA	5.622
mon.27.78abd	3.004	2.037	8.885	2.244	3.188
pil.27.8c9a	4.761	-0.823	-0.153	-1.346	4.461
ple.27.420	2.199	0.884	1.740	-1.038	2.817
sol.27.4	NA	11.410	50.087	3.746	21.180
whb.27.1-91214	6.940	9.241	3.166	1.211	NA

References

- Hilton-Taylor, Craig, and David Brackett. 2000. “2000 IUCN Red List of Threatened Species.”
- Mace, Georgina M, Nigel J Collar, Kevin J Gaston, CRAIG Hilton-Taylor, H Resit Akçakaya, NIGEL Leader-Williams, Eleanor Jane Milner-Gulland, and Simon N Stuart. 2008. “Quantification of Extinction Risk: IUCN’s System for Classifying Threatened Species.” *Conservation Biology* 22 (6): 1424–42.